

Adding year effects

a4a Training Course — Module 2.4

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A year effect on F

The last set of residuals showed a clear pattern **across years** — more positive early in the series, more negative in the recent years. As for age, we answer it with a factor.

```
fit05 <- sca(stk, idx,  
            fmodel = ~factor(age) + factor(year),    # <- year added  
            qmodel = list(~1),  
            srmodel = ~1, vmmodel = list(~1, ~1), n1model = ~1)
```

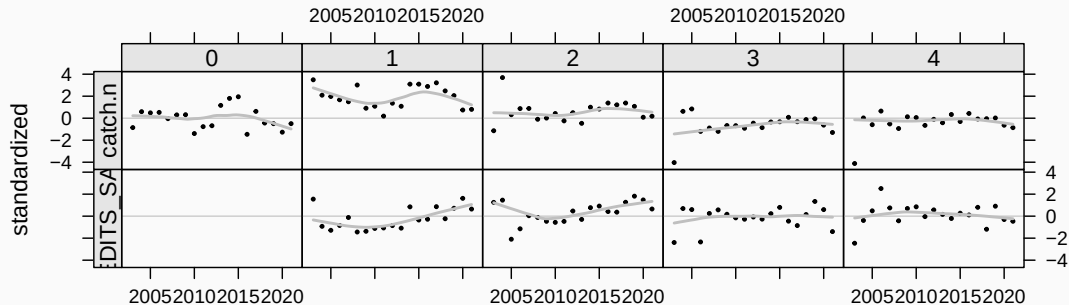
💡 And the scale is pinned down

[1] 0.63 2.75

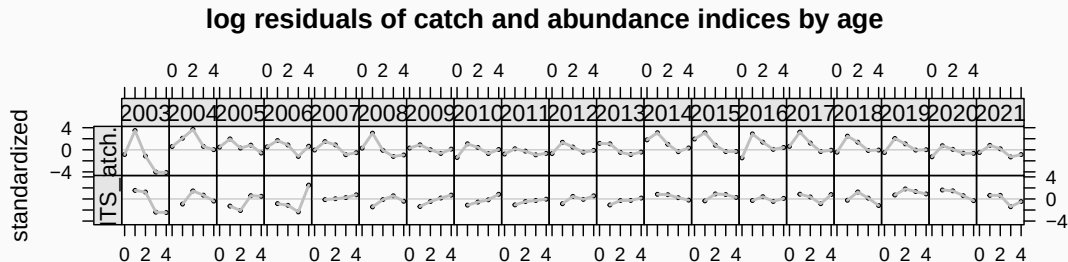
$\bar{F}$ is a real number again, not 10^{-15} . Letting F move through time is what fixes the degeneracy from Module 2.3.

Year effect on F — residuals by year

log residuals of catch and abundance indices by year



Year effect on F — residuals by age



i Not finished

Much less pattern than before — but the survey catchability still looks like it has a year trend, and recruitment, N_1 and the variances are all still constants.

Why we do not add year effects to q

A well-designed scientific survey should see abundance change because **abundance changed**.



Modelling a year trend in q

...attributes that change to the **survey design** instead, and removes it from the abundance. Think carefully before you do it.

The exception is a **commercial CPUE**, where changes in selectivity or fishing behaviour really can enter the index — although standardisation should have dealt with the technical part.

The formulas are the same ones: period breaks, trends, and so on.

The initial population, N1

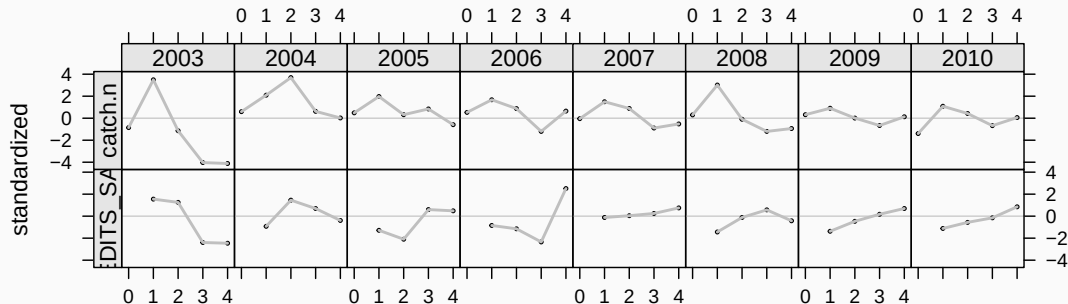
The first year has no earlier data to reconstruct its cohorts from, so `n1model` sets the numbers at age in year 1.

```
fit06 <- sca(stk, idx,  
            fmodel = ~factor(age) + factor(year),  
            qmodel = list(~factor(age)),  
            srmodel = ~1, vmmodel = list(~1, ~1),  
            # <- was ~1  
            n1model = ~factor(age))
```

It affects the **lower triangle** of the population matrix — the early years — and the catches there. So zoom into the start of the series to see it.

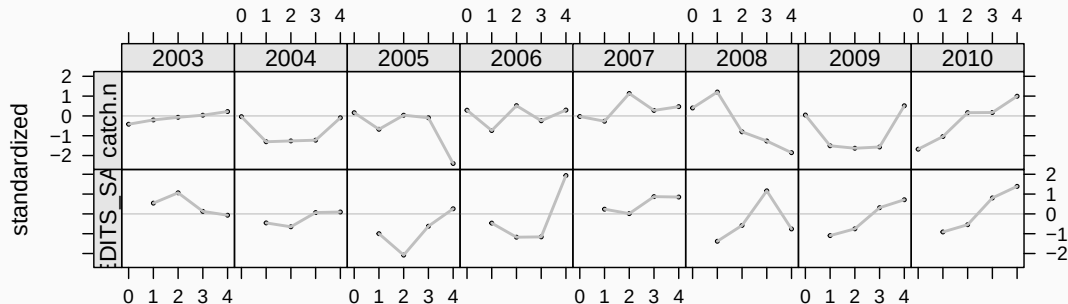
N1 as an intercept: 2003–2010 by age

log residuals of catch and abundance indices by age



N1 with an age effect: 2003–2010 by age

log residuals of catch and abundance indices by age



Catch and survey residuals both improve considerably. The following years improve too, to different degrees.

The stock-recruitment submodel

Constant recruitment is a strong assumption. Let it vary over time:

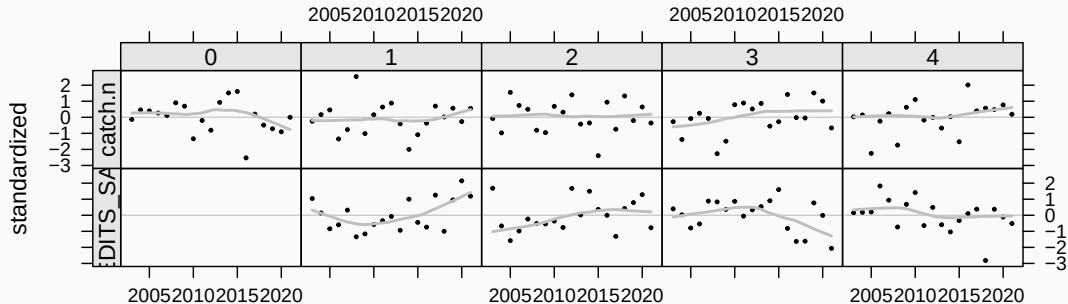
```
fit07 <- sca(stk, idx,  
            fmodel = ~factor(age) + factor(year),  
            qmodel = list(~factor(age)),  
            # <- was ~1  
            srmodel = ~factor(year),  
            vmodel = list(~1, ~1), n1model = ~factor(age))
```

Why it matters everywhere

Unmodelled recruitment variability does not stay put: it was being picked up as a **trend in the survey catchability**, and cohort dynamics carried it into the other ages.

Recruitment varying — residuals by year

log residuals of catch and abundance indices by year



The year trends present in the previous models are gone.

Summary

- A year effect on F **breaks the scale degeneracy** from Module 2.3
- Do **not** put year trends in q for a scientific survey — think first
- `n1model` fixes the **early years**, propagating along cohorts
- `srmodel` lets recruitment vary; unmodelled, it contaminates q and the catch
- Each step was driven by **what the residuals showed**, one change at a time

Exercise

Open `handson_M2.4_year_effects.R`.

1. Add the year effect and check `fbar` is sensible again
2. Add `n1model`, and compare the early years before and after
3. Add `srmodel`, and watch the year trends leave the survey residuals